2023

AUSTRALIAN MATHEMATICS COMPETITION

AMC.

AUSTRALIAN MATHS TRUST

Level B
Upper Primary
Grades **5–6**

DATE

23 September

TIME ALLOWED

60 minutes

Instructions and Information 考试说明

1. You must maintain silence at all times.
   考试过程中请保持安静。
2. Do not open the paper until told to do so.
   收到监考老师指示前，请勿打开试卷。
3. Mobile phones are not allowed. If you have one you must give it to a supervisor now.
   禁止使用手机。如果携带手机，请立刻交给监考老师。
4. You must not leave your seat during the competition.
   考试期间禁止离开座位。
5. You may not borrow equipment without a supervisor's permission.
   未经监考老师许可，禁止向他人借用工具。
6. If you wish to leave the room a supervisor must accompany you.
   如果想离开考场，必须有监考老师陪同。
7. If you have any other questions, raise your hand and wait for a supervisor.
   如果有其他问题，请举手示意监考老师。
8. You are allowed a working time of 60 minutes. There is no extra reading time.
   本场考试时长为 60 分钟，没有额外的读题时间。
9. There are no penalties for incorrect answers. You should attempt all questions.
   答错不扣分，请尝试回答所有问题。
10. The questions have been thoroughly checked. Each question stands as written and no further explanation can be provided.
    本场考试的所有问题已仔细核查，不提供任何口头解释。
11. Diagrams are NOT drawn to scale. They are intended only as aids.
    图形未按比例绘制，仅供参考。
12. You may use calculators and printed language dictionaries.
    允许使用计算器和纸质语言词典。

Important: No student may sit the competition more than once or sit more than one division of the competition. The AMT conducts integrity checks on competition results and reserves the right to withhold results or disqualify students if plagiarism or duplicate sittings are suspected.

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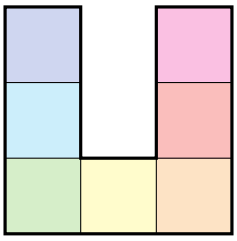
Upper Primary Division

Questions 1 to 10 are worth 3 marks each.

1-10 题，每题 3 分

1. This shape is made from 7 squares, each 1 cm by 1 cm. What is its perimeter?

右侧图形由 7 个 1 cm ×1 cm 的正方形组成。请问这个图形的周长是多少？



(A) 7 cm (B) 12 cm (C) 14 cm (D) 16 cm (E) 28 cm

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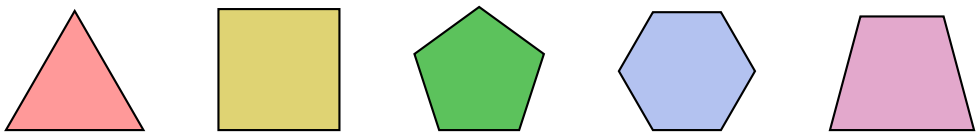
2. There are five shapes here. How many are quadrilaterals?

以下五个图形中，请问有多少个是四边形？



(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

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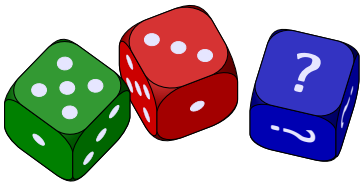
3. In a board game, Nik rolls three standard dice, one at a time. He needs his three rolls to add to 12. His first two dice rolls are 5 and 3. What does he need his third roll to be?



一个棋盘游戏中，小尼投掷三枚标准的骰子，每次掷出一枚。他需要三次掷出的点数之和为 12。

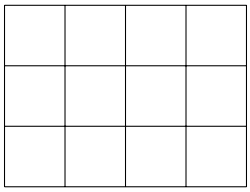
他前两次掷出的点数分别是 5 和 3，请问小尼第三次需要掷出的点数是多少？

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

---

4. In this diagram, how many of the small squares need to be shaded for the large rectangle to be one-quarter shaded?

要使得图中阴影面积占大长方形面积的四分之一，请问需要给多少个小正方形涂上阴影？



(A) 2 (B) 3 (C) 4 (D) 6 (E) 12

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5. Petra left for school at 8:51 am. She got to school at 9:09 am.

How long did it take Petra to get to school?

小潘在上午 8:51 出发去学校，上午 9:09 到达学校。

请问小潘去学校花了多长时间？

(A) 9 minutes 9 分钟 (B) 10 minutes 10 分钟 (C) 18 minutes 18 分钟

(D) 42 minutes 42 分钟 (E) 1 hour 1 小时

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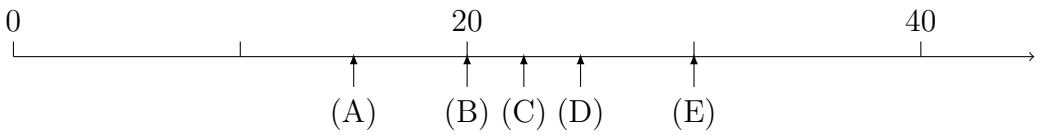
6. Which letter marks where 25 is on this number line?

请问哪个字母表示下方数轴上数字 25 的位置？



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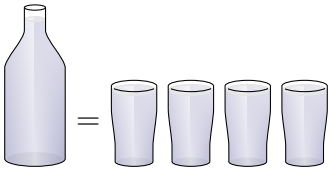
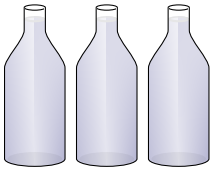
7. This bottle holds 4 glasses of water.

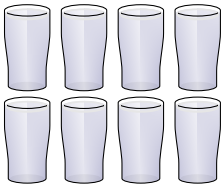
一个瓶子可以装 4 杯水。

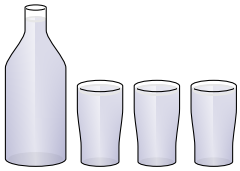


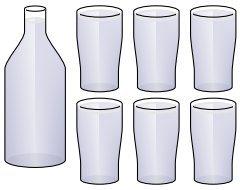
Which one of the following holds the most water?

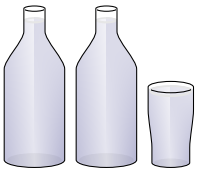
以下哪项装的水最多？

(A)


(B)


(C)


(D)


(E)


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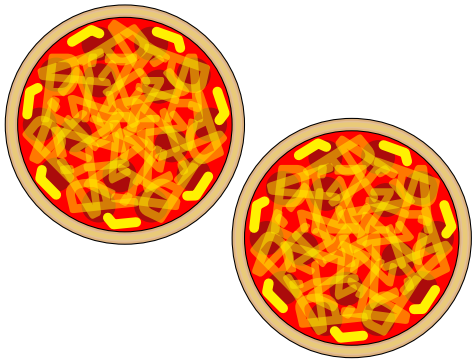
8. Two pizzas are shared equally between 3 students.

What fraction of a whole pizza does each student get?

将两个披萨平均分给 3 个学生。请问每个学生可以得到一整个披萨的几分之几？



(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$ (E) $\frac{3}{4}$

---

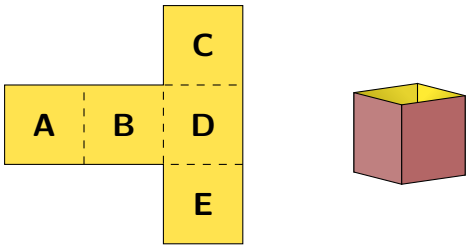
9. A piece of card is cut out and labelled as shown in the diagram.

It is folded along the dotted lines to make a box without a top.

Which letter is on the bottom of the box?

将一张卡片裁剪成如图所示的形状，并标记上字母，然后将其沿着虚线折叠成一个没有顶部的盒子。

请问哪个字母位于盒子底部？



(A) A (B) B (C) C (D) D (E) E

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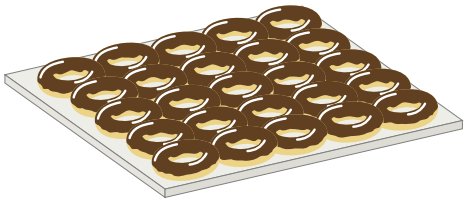
10. Doughnuts come in bags of 3 and boxes of 8.

I bought exactly 25 doughnuts for my party.

What do I get when I add the number of boxes I bought and the number of bags I bought?

甜甜圈以每袋 3 个和每盒 8 个的方式进行出售。我为聚会恰好购买了25 个甜甜圈。如果将我买的甜甜圈的盒数和袋数相加，请问得到的数字是多少？



(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

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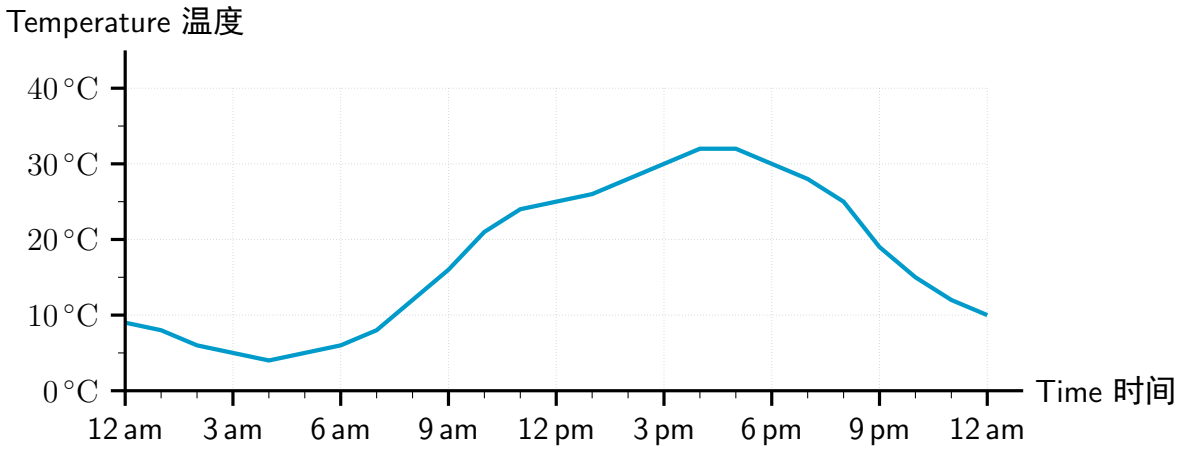
Questions 11 to 20 are worth 4 marks each.

11-20 题，每题 4 分

11. This line graph shows the temperature each hour during a day.

以下折线图显示了一天中每小时的温度。



Roughly for how long was the temperature above 20 °C?

请问温度高于 20 °C 的时间大约有多长？

(A) 7 hours 7 小时

(B) 8 hours 8 小时

(C) 9 hours 9 小时

(D) 10 hours 10 小时

(E) 11 hours 11 小时

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12. VIV takes her three children, HANNAH, OTTO and IZZI, out shopping. Each is wearing a t-shirt with their name on the front in capital letters.

When they stand in front of the shop mirror, which names appear the same in the reflection as on the shirts?

VIV 带着她的三个小孩 HANNAH、OTTO 和 IZZI 去购物。每个孩子穿着印有自己名字的 T 恤，名字用大写字母印在 T 恤正面。当他们站在商店镜子面前时，请问哪些名字在镜子中的图像与衬衫上的名字相同？

(A) VIV and OTTO　　VIV 和 OTTO

(B) VIV, OTTO and IZZI　　VIV、 OTTO 和 IZZI

(C) VIV, HANNAH and IZZI　　VIV、 HANNAH 和 IZZI

(D) HANNAH and OTTO　　HANNAH 和 OTTO

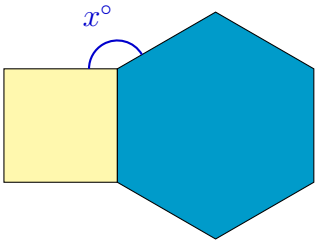
(E) All four of them 四个名字都相同

---

13. This regular hexagon has angles of 120° and the square has angles of 90°.

What is the angle $x°$ in the diagram?

图中正六边形的内角为 120°，正方形的内角为 90°。请问图中角 $x°$ 的度数是多少？



(A) 90°　　(B) 120°　　(C) 135°

(D) 150°　　(E) 180°

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14. Syed's mother had some money to share with her family.

She gave one-quarter of her money to Syed.

Then she gave one-third of what was left to Ahmed.

Then she gave one-half of what was left to Raiyan.

She was left with \$15, which she kept for herself.

How much money did Syed's mother have to start with?

小赛的妈妈要分一些钱给家人。

她把四分之一的钱给了小赛，

把剩下的钱的三分之一给了小艾，

然后她再把剩下的钱的一半给了小莱，

最后还剩下 \$15。

请问小赛的妈妈最开始有多少美元？

(A) \$30 (B) \$45 (C) \$60 (D) \$90 (E) \$120

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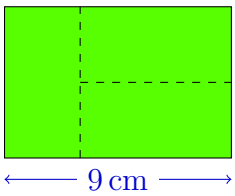
15. The rectangle shown has a side length of 9 cm.

It is divided into 3 identical rectangles as shown.

What is the area, in square centimetres, of the original rectangle?



如图所示，一个长为 9 cm 的长方形被分成三个相同的小长方形。

请问原来长方形的面积是多少平方厘米？

(A) 45 (B) 50 (C) 52 (D) 54 (E) 63

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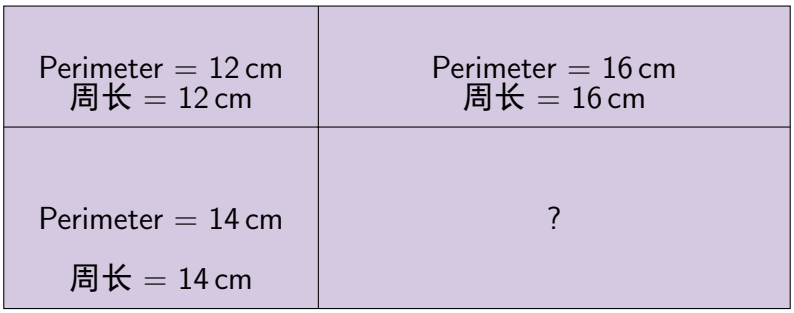
16. This diagram shows a rectangle with a perimeter of 30 cm. It has been divided by 2 lines into 4 small rectangles. Three of the small rectangles have the perimeters shown. What is the perimeter of fourth small rectangle?

如图所示，一个周长为 30 cm 的大长方形被两条直线分成 4 个小长方形，其中三个小长方形的周长已标注出来。请问第四个小长方形的周长是多少？



(A) 10 cm (B) 12 cm (C) 14 cm (D) 16 cm (E) 18 cm

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17. There are 10 questions in a test. Each correct answer scores 5 points, each wrong answer loses 3 points, and if a question is left blank it scores 0 points.

Tycho did this test and scored 27 points. How many questions did Tycho leave blank?

一场测试中有 10 道题目，答对一题得 5 分，答错一题扣 3 分，不答得 0 分。小迪参加了这场测试，得分为 27 分。请问小迪有多少道题目没有作答？

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

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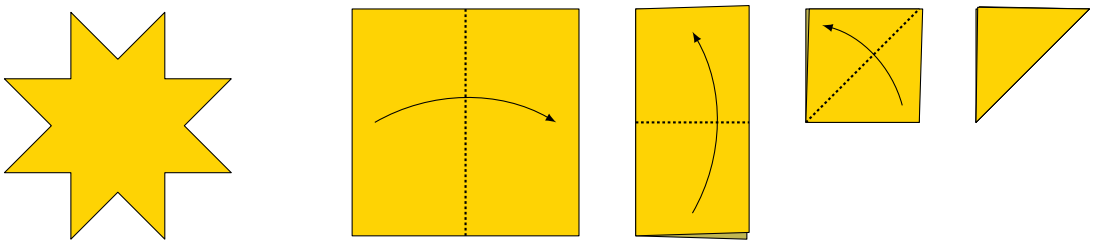
18. Estelle is making decorations shaped like the 8-pointed star shown.

She folds a square of paper to make a triangle with 8 layers as shown.

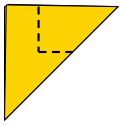
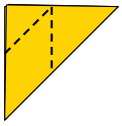
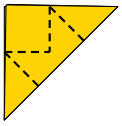
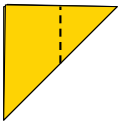
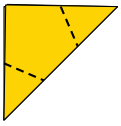
小爱正在制作八角星形状的装饰品。

如图所示，她将一张正方形纸张，折叠成有 8 层的三角形。



How could she cut the triangle so that the unfolded shape is the star?

请问小爱如何裁剪这个三角形，才能使得展开后的形状是八角星形？

(A)  (B)  (C)  (D)  (E) 

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19. Earlier this year Ben said, 'Next year I will turn 13, but 2 days ago I was 10.'

Ben's birthday is

今年早些时候，小贝说："明年我就 13 岁了，但两天前我才 10 岁。"

请问小贝的生日是几月几日？

(A) 1st January 1 月 1 日

(B) 2nd January 1 月 2 日

(C) 29th December 12 月 29 日

(D) 30th December 12 月 30 日

(E) 31st December 12 月 31 日

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20. Peyton, Luka and Dan have 180 stickers in total. Peyton has half as many stickers as Luka. Dan has three times as many as Luka.

How many stickers does Peyton have?

小佩、小鲁和小丹一共有 180 张贴纸。小佩的贴纸数是小鲁的一半，小丹的贴纸数是小鲁的三倍。

请问小佩有多少张贴纸？

(A) 20 (B) 24 (C) 30 (D) 40 (E) 54

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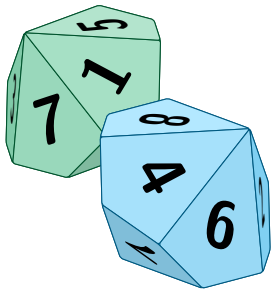
Questions 21 to 25 are worth 5 marks each.

21-25 题，每题 5 分

21. Mrs Graaf invents a game for her students to practise arithmetic. They roll two 10-sided dice to pick two random numbers. Starting at one of the numbers, they keep adding the other number until they reach a 3-digit number.



Ian rolls a 5 and an 8. If he chooses to start with 5 and then add 8 again and again, his list is 5, 13, 21, . . . , 93, 101. If he chooses to start with 8 and add 5, his list is 8, 13, 18, . . . , 98, 103.

On Nara's turn, she makes a list that ends with 107.

What pair of numbers could she have rolled?

格拉夫人为她的学生设计了一个练习计算的游戏。学生们投掷两枚 10 面骰子得到两个随机数，然后选择其中一个数开始，反复加上另一个数，直到得出一个三位数。小伊掷出一个 5 和一个 8。如果他选择从 5 开始，然后反复加上 8，得到的数列是 5，13，21，⋯，93，101。如果他选择从 8 开始，反复加上 5，得到的数列是 8，13，18，⋯，98，103。在小娜的回合中，她得到的数列以 107 结束。

请问她可能掷出了哪对数字？

(A) 4 and 8　4 和 8　(B) 5 and 7　5 和 7　(C) 3 and 4　3 和 4

(D) 6 and 9　6 和 9　(E) 3 and 8　3 和 8

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22. At a school concert, the tickets cost \$20 per adult and \$2 per child. The total paid by the 100 people who attended was \$920. How many were children?

在一场学校音乐会上，成人票价为 20 美元，儿童票价为 2 美元。参加音乐会的 100 人支付的总金额为 920 美元。请问其中有多少名是儿童？

(A) between 25 and 35　　25 名至 35 名

(B) between 35 and 45　　35 名至 45 名

(C) between 45 and 55　　45 名至 55 名

(D) between 55 and 65　　55 名至 65 名

(E) between 65 and 75　　65 名至 75 名

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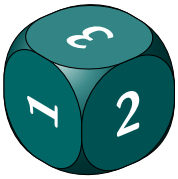
23. Meena has a standard dice, with each pair of opposite faces adding to 7. At first, the three faces she can see add to 6, as shown.



She holds the dice between a pair of opposite faces and rotates it 180°, keeping these opposite faces facing the same direction. She puts the dice back down and adds up the three faces she can now see.

What is the smallest possible total she could get?

小美有一枚标准骰子，相对面的点数之和为 7。如图所示，她现在可以看到的三个面的点数之和是 6。现在她抓住骰子上一对相对的面，旋转180°，保持相对面的方向不变。然后将骰子放下，并将能看到的三个面的点数相加。

请问得到的和的最小可能值是多少？

(A) 6　　(B) 8　　(C) 10　　(D) 12　　(E) 14

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24. I have 4 whole numbers that add up to 98.

If I were to add 6 to the first number, subtract 6 from the second number, multiply the third number by 6 and divide the fourth number by 6, the four answers would all be the same.

What is the sum of the largest two of my original four numbers?

我有四个整数，它们的和为 98。

将第一个数字加上 6，第二个数字减去 6，第三个数字乘以 6，第四个数字除以 6，得到的四个结果都相等。

请问最初的四个数字中，最大的两个数字之和是多少？

(A) 72　　(B) 86　　(C) 88　　(D) 90　　(E) 94

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25. When I ride my bike at 20 kilometres per hour, each wheel turns at 2 revolutions per second.

When I ride 1 kilometre, how many revolutions does each wheel make?

当我以每小时 20 公里的速度骑自行车时，每个车轮每秒转 2 圈。当我骑行 1 公里时，请问每个车轮转了多少圈？

(A) 40 (B) 240 (C) 320 (D) 360 (E) 420

For questions 26 to 30, colour in the bubbles on the answer sheet to record whole-number answers from 0 to 999. Questions 26 to 30 are worth 6, 7, 8, 9 and 10 marks respectively.

26 题至 30 题的答案为 0 至 999 的整数，请正确填涂在答题卡上。第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，第 29 题占 9 分，第 30 题占 10 分。

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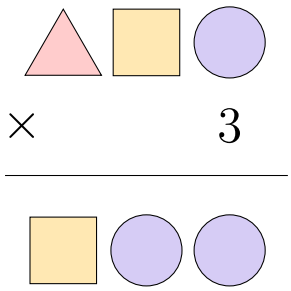
26. In this puzzle, △, □ and ○ represent different nonzero digits.

What is the three-digit number △□○?

在右侧谜题中，△、□ 和 ○ 表示不同的非零数字。

请问这个三位数 △□○ 是多少？



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27. Li attempted to multiply a single-digit number by 36, but he accidentally multiplied by 63 instead. His answer was 189 larger than the correct answer.

What was the correct answer to the multiplication?

Li 尝试将一个一位数乘以 36，但他不小心将这个一位数乘以了 63。最终得到的结果比正确结果大 189。请问这个乘法计算的正确结果是多少？

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28. Using 9 out of the 10 possible digits Safia writes 3 numbers, each between 100 and 999. She adds her 3 numbers together. What is the smallest possible sum?

小萨从 10 个可能的一位数中选出 9 个一位数，**组成** 3 个数，每个数都介于100 到 999 之间。将这 3 个数相加，请问得到的最小可能值是多少？

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29. Yifan has a construction set consisting of red, blue and yellow rods. All rods of the same colour are the same length, but differently coloured rods are different lengths. She wants to make quadrilaterals using these rods.

小伊有一个含有红色、蓝色和黄色小棒的组装套件。颜色相同的小棒长度相同，颜色不同的小棒长度也不同。她想用这些小棒拼成四边形。

- When she uses two red, one blue and one yellow rod, the perimeter of the quadrilateral is 36 cm.
- When she uses two blue, one red and one yellow rod, the perimeter is 35 cm.
- When she uses two yellow, one blue and one red rod, the perimeter is 33 cm.

- 使用两根红色、一根蓝色和一根黄色小棒拼成的四边形周长是 36 cm。
- 使用两根蓝色、一根红色和一根黄色小棒拼成的四边形周长为 35 cm。
- 使用两根黄色，一根蓝色和一根红色小棒拼成的四边形周长为 33 cm。

What number do you get when you multiply the lengths of one red rod, one blue rod and one yellow rod?

将一根红色、一根蓝色和一根黄色小棒的长度相乘，请问得到的数是多少？

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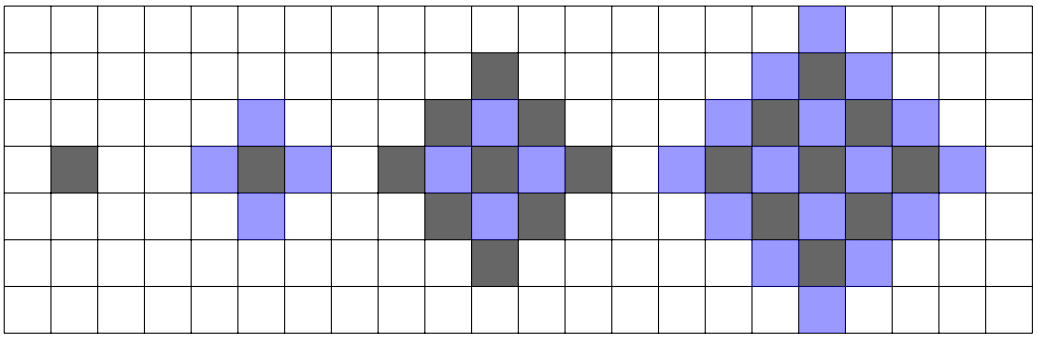
30. Janus is making patterns using square tiles. Each pattern is made by copying the previous pattern, then adding a tile to every grid square that shares an edge with the copied pattern.

小简正在用正方形瓷砖制作图案。每个图案都是先复制上一个图案，再将与复制图案共用一条边的每个方格中添加一块瓷砖。



Pattern 1 图案 1　Pattern 2 图案 2　Pattern 3 图案 3　Pattern 4 图案 4

His last pattern is the largest one that can be made with fewer than 1000 tiles. How many tiles are in this last pattern?

他的最后一个图案是使用少于 1000 块瓷砖可以制作出来的最大图案。

请问最后一个图案有多少块瓷砖？

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